

Ke Li

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Pronouns: she/her

Research Interests

My research lies at the intersection of AI and decision science. I pursue two complementary streams:

- **Data-Driven Optimization:** developing data-driven machine learning algorithms to support robust decision-making under uncertainty.
- **Human-AI Collaboration in Scientific Process:** studying how machine learning and generative AI can augment the scientific process, particularly in theory building.

Across both streams, I develop computationally rigorous and decision-relevant methods that advance hybrid human-AI intelligence.

Education

INSEAD

France

Ph.D. Candidate in Decision Sciences (AI/ML + Optimization)

09/2023 - 06/2028

Advisor: Spyros Zoumpoulis

The Chinese University of Hong Kong, Shenzhen

China

B.Eng (Honors) in Computer Science and Engineering

09/2018 - 06/2022

Alumni Interview Artical ([CN](#) version/[EN](#) version)

National College Entrance Examination (“Gaokao” in China)

Henan, China

Total score: 660, top 0.1% (ranking 1000 over around 1 million peers in Henan Province)

Honors & Awards

Teaching Excellence Award in PhD Tutorials, INSEAD	2025
“Stars of Tomorrow” Award of Excellent Intern (Top 10%), Microsoft Research Asia [link]	2022
ExploreCSR Computing Research Award, Google Research & UTRGV [link]	2021
Dean’s List recipient, CUHK-Shenzhen	2019-20
Undergraduate Research Awards, CUHK-Shenzhen	2020
Undergraduate Student Teaching Fellows, CUHK-Shenzhen	2020
China National Encouragement Scholarship, Ministry of Education of China	2019
Undergraduate Admission Fellowships, CUHK-Shenzhen	2018-22

Publications - [See Google Scholar](#)

1. Zhou, B., Li, K., Jiang, J., & Lu, Z. Learning from Visual Observation via Offline Pretrained State-to-Go Transformer. *Advances in Neural Information Processing Systems (NeurIPS, Top-tier AI/CS/ML conference)*, 2023.
2. Dong, J., Li, K., Li, S., & Wang, B. Combinatorial bandits under strategic manipulations. In *the Fifteenth ACM International Conference on Web Search and Data Mining (WSDM, Top-tier data mining conference)*, 2022.

3. Liu, Y., Li, K., Huang, Z., Li, B., Wang, G., & Cai, W. EduChain: a blockchain-based education data management system. In *Blockchain Technology and Application: Third CCF China Blockchain Conference*, 2021.

Work in Progress

- **A Human-AI Collaborative Framework for Theory Building** - Joint work with [Eric Uhlmann](#), [Phanish Puranam](#), and [Spyros Zoumpoulis](#).

Presented at:

- Selected Presentation, AI Plus Management Doctoral Consortium, London UK, 2025 - [link](#)
- Invited Session, INFORMS Annual Meeting, Atlanta USA, 2025 - [link](#)
- **Optimizing for Decisions: A Decision-Aware Learning Approach** - Joint work with [Sandeep Chandukala](#), [Ernst Osinga](#), and [Spyros Zoumpoulis](#).

Contributions to Open-Source Community

- **Microsoft Research Asia** — Contributed to Microsoft's AutoML toolkit [NNI](#) (14,300+ GitHub stars), including transformer model-compression modules now integrated into both internal and external AutoML pipelines.
- **SenseTime (OpenDI Lab)** — Contributed 1,600+ lines of code to the [DI-engine](#) reinforcement-learning framework (3,600+ GitHub stars), improving algorithmic stability and large-scale training utilities.

Working Experiences

These full-time and internship experiences provided me with deep technical expertise in machine learning, reinforcement learning, and large-scale experimental systems.

Beijing Academy of Artificial Intelligence
Reinforcement Learning Researcher (Full-time)

Beijing
12/2022 - 07/2023

- Designed and trained intelligent agents capable of learning directly from high-dimensional visual observations in complex 3D game environments such as Minecraft.
- Developed novel architectures combining convolutional encoders with deep reinforcement learning algorithms (e.g., PPO, DQN), enabling agents to acquire long-horizon skills like navigation, object manipulation, and environment interaction.
- Led extensive experiments to validate model performance, and collaborated closely with researchers from Tsinghua and Peking University.
- Our research paper ([website](#)) is accepted by NeurIPS 2023!

Microsoft Research Asia
Research Intern, Innovation Engineering Group, Award of Excellent Intern

Beijing
01/2022 - 06/2022

SenseTime (OpenDI Lab)
Reinforcement Learning Intern

Shenzhen
08/2021 - 12/2021

Teaching Assistant

INSEAD

Probability and Statistics, PhD Course (Winning Teaching Excellence Award)	2024.10 - 2024.12
FAIM - Foundations of AI for Managers, MBA Course	2025.01 - 2025.02
FAIM - Foundations of AI for Managers, MBA Course	2025.05 - 2025.06

The Chinese University of Hong Kong

MAT1010 Calculus I (Undergraduate Student Teaching Fellowship)	Fall 2019-2020
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Academic Services

Session Chair (co-chair with Spyros Zoumpoulis), INFORMS Annual Meeting <i>Session Title: Estimation and Optimization in Data-Driven Decisions</i>	2025
Reviewer , NeurIPS (Top-tier conference in CS/ML/AI)	2024

Patents

- Personnel exemption aided decision making method and system based on historical big data. China Patent CN113673943A. [\[link\]](#)